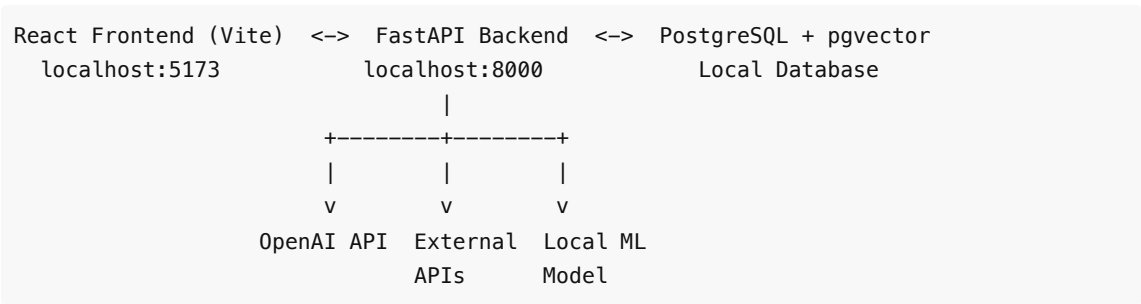


AI Capability Mapping - Technical Architecture Documentation

Project Overview

System Type: RAG-based Academic Q&A System
Tech Stack: React + FastAPI + PostgreSQL + OpenAI API
Data Scope: UNSW staff information and academic publications (publicly available data only)

System Architecture



Core Components

Layer	Technology	Description
Frontend	React 19 + Vite	SSE streaming for real-time AI responses
Backend	FastAPI + Uvicorn	RESTful API with streaming support
Database	PostgreSQL 14+ + pgvector	Structured data + 1536-dim vector embeddings
Search Engine	BM25 + Vector Search + Reranker	Hybrid retrieval with RRF fusion
AI Model	OpenAI gpt-4o-mini	Answer generation + text embeddings
Local Model	Sentence-Transformers	Reranker (~600MB memory)

External Service Dependencies

Service	Purpose	Cost	Risk Level
OpenAI API	LLM generation + embeddings	Paid (per token)	Critical Dependency

Infrastructure Requirements

Resource	Minimum Requirement
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Memory	2GB+ (Reranker model uses approximately 600MB)
CPU	2 cores+
Storage	10GB+ (database + models)
Python	3.10+
PostgreSQL	14+ (pgvector extension required)
Network	Access to OpenAI API

Data Compliance and Privacy

Data Type	Source	Sensitivity Level
Staff Information	UNSW public website	Low (public data)
Publication Data	OpenAlex & academic APIs	Low (public data)
Vector Embeddings	Generated by OpenAI	Low

No Privacy Concerns - The system only processes publicly available academic information. No personal or sensitive data is stored or processed.